RUIQI DING (22 YO)

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RESEARCH EXPERIENCE

Reinforcement Learning Nanobot

Reiner Kree group, University of Göttingen

Apr 2019 – Aug 2019

- We study the Statistical Physics of disordered and complex systems such as: biomembranes, cell cultures, neural networks and biological neurons.
- Specifically, we successfully trained nanobots to navigate through non-observable complex fluids such as human tissues using **deep reinforcement learning**. The fluid environment was characterised by Markovian transition matrices so that the nanobot undergoes random walks at each time step.
- tesla-cat/Reinforcement-Learning-Nanobot

FPGA Trapped Ion Control

Dzmitry Matsukevich (h-index: 30) group, CQT

Aug 2019 – Dec 2019 dzmitrylab.quantumlah.org

- We use trapped Ytterbium ions to study quantum information processing and the implementation of heat machines in the quantum regime, as well as precision measurements with atomic and molecular ions.
- Specifically, I worked on FPGA programming using the state of the art Cyclone V SoC FPGAs. The FPGA is used to generate sequencing signals to control the laser pulse that drives trapped ions, it is also used for photon counting. In addition, I developed a full-stack framework that handles the communication between FPGA hardware and Python programs.
- tesla-cat/FPGA-Trapped-Ion-Control

Density Potential Functional Theory (Bachelor Thesis)

Berge Englert (h-index: 51) group, CQT

Aug 2019 – May 2020

- We study low-dimensional ultracold dilute gases of fermions and bosons, the band structure, transport properties, and finite-temperature effects of many-body systems in periodic lattices.
- Specifically, I worked on the Density Potential Functional Theory (DPFT) established by Julian Schwinger (Nobel Prize in Physics, 1965) and his postdoc Berge Englert. Its application on spin polarized Fermi gas with magnetic dipole-dipole interaction was studied under the Thomas Fermi approximation in both position and momentum space. The code used the **Machine Learning** library PyTorch to achieve multi-GPU acceleration.
- tesla-cat/PyDPFT

Reinforcement Learning Quantum Control

Yvonne Y. Gao (MIT TR35) group, CQT

Jul 2020 – Ongoing quantumcrew.org

MY DREAM

Theoretically: to understand the physics and mathematics of consciousness and intelligence. Practically: to develop clinically available non-invasive brain-computer interfaces. If anything ends up stopping me from chasing my dream, it will not be my poor family background but my laziness and excuses.

EDUCATION

M.Sc. in Physics

University of Göttingen, Germany

Apr 2019 – Aug 2019

- GPA: mine is 1.4 (Very good in German system), best is 1.0, **pass** is 4.0
- Transcript: 3-uni-goettingen-grades.pdf
- This was supposed to be a bachelor exchange program, but I liked challenges, so I took master courses in Biophysics.

B.Sc. in Physics with Honours (Highest Distinction)

National University of Singapore

Aug 2016 – May 2020

- GPA: mine is 4.52 (Highest Distinction), best is 5.0, worst is 0
- Degree Scroll: 2-NUS-Degree.pdf
- Transcript: 2-NUS-Transcript.pdf
- Thesis (grade: 5.0): Density Potential Functional Theory in position and momentum space and its implementation using the Machine Learning library PyTorch.

AWARDS

China-Singapore intergovernmental scholarship
Only 90 quota each year in the entire China, only available to best students in best high schools.

National University of Singapore Science and Technology Undergraduate Scholarship
Continuation of the above, awarded to those who pass one year of observation period.

- We design novel quantum circuits, develop robust hardware for quantum computers, probe quantum effects at 10mK, and build test-beds for quantum algorithms.
- Currently, I am working on quantum optimal control theory using the state of the art **deep reinforcement learning** algorithms.
- Qcrew/Reinforcement-Learning-Quantum-Control

PROJECTS

Lab Kitten

- Lab Kitten is a toolkit for labs / experiments using real-time communication technology, it enables researchers to monitor lab data and control lab instruments conveniently no matter where they are, for example at home or on a bus.
- Specifically, it uses WebRTC technology, which is the real-time technology of highest performance and has been widely used in live-streaming, video chat applications etc.
- lab-kitten/lab-kitten

FPGA-IoT Musical Tesla Coil

- The goal is to build a Musical Tesla Coil using two advanced technologies: IoT and FPGA.
- A front-end UI was written in Javascript, it parses MIDI music files to extract frequencies and volumes, then sends them to the ESP32 IoT module that runs a websocket server, which in turn sends SPI and PWM signals to control the FPGA. The FPGA handles controlled phase shifting to drive the IGBT Full bridge inverter, which in turn drives the Tesla Coil's LC resonator to produce high-voltage plasma music. All PCBs were designed and soldered by myself.
- tesla-cat/FPGA-ESP32-Musical-Tesla-Coil
- 1-musical-tesla-coil.mp4

CODING SKILLS (SELF-TAUGHT)

Python | PyTorch | JavaScript | React.js | C++ | Arduino | Verilog | Quartus | MATLAB | Mathematica

OTHER SKILLS

Physics and Math (analytical) | Altium (electronics design, self-taught) | SolidWorks (mechanical design, self-taught)

China's nationwide high school physics competition third prize

China's nationwide middle school math competition second prize

HUAWEI High School Science Fair best performance award

REFEREES

Prof. Berge Englert
@ PI at CQT
cqtebg@nus.edu.sg

- Prof. Englert is one of the funding figures of Quantum Optics (together with Marlan Scully), he was postdoc for Julian Schwinger (Nobel Prize in Physics, 1965), and the Scientific Secretary of the Julian Schwinger Foundation.
- He was the supervisor for my bachelor's thesis and my professor for many other modules.

Prof. Helmut Grubmüller
@ Director of Max Planck Institute for Biophysical Chemistry
hgrubmu@gwdg.de

- Prof. Grubmüller is the world's leading scientist in Theoretical and Computational Biophysics (ranked No.1 on Google Scholar, h-index: 76)
- He was my professor for modules *Research Seminar on Biophysics / Physics of Complex Systems* (my grade: 1.7) and *Biomolecular Physics and Simulations* (my grade: 1.3)

Prof. Reiner Kree
@ University of Göttingen
kree@theorie.physik.uni-goettingen.de

- Although he doesn't have a Google Scholar profile, Prof. Kree is an outstanding German physicist. He used to be the Dean of the Faculty of Physics and Vice President of the University of Göttingen. This is extraordinary considering the sacred status of University of Göttingen in Math and Physics.
- He was my professor for the module *Advanced Topics in Biophysics / Physics of complex systems* (my grade: 1.0, best) and previously mentioned Reinforcement-Learning-Nanobot project.